

Business Context

In the rapidly evolving landscape of e-commerce, customer reviews play a pivotal role in shaping the perception of products and influencing purchasing decisions. All businesses recognize the significance of understanding customer sentiment and extracting valuable insights from the vast pool of customer reviews to drive decisions. The inability to actively track customer feedback can result in significant challenges for businesses. Negative sentiments expressed by customers become crucial to track and resolve as they can lead to several detrimental effects, including:

- **Customer Churn:** Unaddressed negative sentiment can lead to customer dissatisfaction and the potential loss of loyal customers. This may result in customers opting for alternative platforms for their future purchases, impacting long-term revenue and market share.
- **Reputation Damage:** The persistence of negative sentiment, if not effectively managed, can tarnish the reputation of the business. Potential customers may be dissuaded from making purchases on the platform due to concerns raised in negative reviews, affecting brand loyalty and trust.
- **Financial Implications:** Negative sentiment can lead to a decline in sales, impacting profitability. Discouraged by negative feedback, potential customers may choose competitors, resulting in dire financial repercussions for the company.

Problem Definition

A rising e-commerce platform in the space of electronic gadgets captures customer feedback through different channels, such as product review forms, surveys, and more. In addition, they also capture customer feedback from social media platforms. After witnessing a 200% growth in their customer base in the last three years, they are now experiencing a substantial upswing in the number of customer feedbacks received, with a noteworthy increase of ~25% over the past six months.

As they were previously monitoring customer feedback manually, this surge in customer feedback presents a multifaceted challenge for the business as the volume of reviews has surpassed the manual capacity to track and analyze effectively. Recognizing the need for a proactive and scalable solution, the organization seeks to employ artificial intelligence (AI) to aid them. The primary objective is to develop an AI solution that can automatically assess and categorize customer sentiments as negative, positive, or neutral by parsing and analyzing product reviews. As a Data Scientist, you are provided with the reviews shared by customers regarding their experience with the product along with the manually labeled sentiments of the reviews and asked to analyze the data and build a predictive model that can parse customer reviews and predict the sentiment of the customer.

Data Dictionary



Problem Statement - Product Reviews Sentiment Analysis

sentiment: sentiment associated with the product review, indicating whether the review expresses a positive, negative, or neutral sentiment

